Tutorial self-text questions

PE/PG/NG 154

Year 1. Analytical and Physical Chemistry

1. self-testWhat is the normality of each of the following acid or base solutions?

(a) 0.35 M HCl;

(b) 0.35 M H_2SO_4;

(c) 0.35 M H_3PO_4;

(d) 0.35 M NaOH.

2. What is the normality of each of the following acid or base solutions?

(a) *0.105 M* $Ca(OH)_2$;

(b) 0.105 M $Al(OH)_3$;

(c) 0.105 M HNO_3;

(d) 0.105 M H_2S.

3. What is the normality of a solution that contains *7.08* g of H_3PO_4 in *185* mL of solution?

4. Calculate the molarity and the normality of a solution that was prepared by dissolving *24.2* g of sodium hydroxide in enough water to make *4000* mL of solution.

5. Calculate the molarity and the normality of a solution that contains *19.6* g of arsenic acid, H_3PO_4, in enough water to make *500*. mL of solution.

6. What are the molarity and normality of a sulfuric acid solution that is *19.6%* H_2SO_4 by mass? The density of the solution is *1.14* g/mL.

7. A *25.0* mL sample of *0.206* M nitric acid solution required *35.2* mL of barium hydroxide solution for neutralisation. Calculate the molarity of the barium hydroxide solution.

8. An impure sample of $(COOH)_2.2H_2O$ that had a mass of *2.00* g was dissolved in water and titrated with standard NaOH solution. The titration required *38.32* mL of *0.198* M NaOH solution.

Calculate the percent $(COOH)_2$. $2H_2O$ in the sample. Assume that the sample contains no acidic impurities.

9. A *50.0* mL sample of *0.0500* M $Ca(OH)_2$ is added to *10.0* mL of *0.200* M HNO_3. (a) Is the resulting solution acidic or basic?

(b) How many moles of excess acid or base are present?

(c) How many additional mL of *0.0500* M $Ca(OH)_2$ or *0.200* M HNO_3 would be required to completely neutralize the solution?

10. An antacid tablet containing calcium carbonate as an active ingredient required *24.5* cm^3 of *0.0932* M HCl for complete neutralisation. What mass of $CaCO_3$ did the tablet contain?

11. Butyric acid, whose empirical formula is C_2H_4O, is the acid responsible for the odour of rancid butter. The acid has one ionisable hydrogen per molecule. A *1.000* g sample of butyric acid is neutralised by *54.42* mL of *0.2088* M NaOH solution. What are

(a) the molecular weight and

(b) the molecular formula of butyric acid?

12. What is the molarity of a solution of sodium hydroxide, NaOH, if *36.2* mL of this solution is required to react with *25.0* mL of *0.0513* M nitric acid solution according to the following reaction?

$$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$$

13. What is the molarity of a solution of sodium hydroxide, NaOH, if *36.9* cm^3 of this solution is required to react with *35.2* cm^3 of *0.101* M hydrochloric acid solution according to the following reaction?

$$HCl + NaOH \rightarrow NaCl + H_2O$$

14. Calculate the normality and molarity of an HCl solution if *43.1* cm^3 of the solution reacts with *0.318* g of Na_2CO_3.

$$2HCl + Na_2CO_3 \rightarrow 2NaCl + CO_2 + H_2O$$

15. Calculate the normality and molarity of an H_2SO_4 solution if *40.0* cm^3 of the solution reacts with *0.424* g of Na_2CO_3.

$$H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + CO_2 + H_2O$$

16. Balance the following equations. For each equation, tell what is oxidised, what is reduced, what is the oxidising agent, and what is the reducing agent.

(a) $MnO_4^-(aq) + Br^-(aq) \rightarrow Mn^{2+}(aq) + Br_2(l)$ *[Acidic medium]*

(b) $Cr_2O_7^{2-}(aq) + I^-(aq) \rightarrow Cr^{3+}(aq) + I_2(s)$ *[Acidic medium]*

(c) $MnO_4^-(aq) + SO_3^{2-}(aq) \rightarrow Mn^{2+}(aq) + SO_4^{2-}(aq)$ *[Acidic medium]*

(d) $Cr_2O_7^{2-}(aq) + Fe^{2+}(aq) \rightarrow Cr^{3+}(aq) + Fe^{3+}(aq)$ *[Acid medium]*

17. Balance the following ionic equations. For each equation, tell what is oxidised, what is reduced, what is the oxidising agent, and what is the reducing agent.

(a) $Al(s) + NO_3^-(aq) \rightarrow Al(OH)_4(aq) + NH_3(g)$ *[Basic medium]*

(b) $NO_2(g) \rightarrow NO_3^-(aq) + NO_2^-(aq)$ *[Basic medium]*

(c) $MnO_4^-(aq) + NO_2^-(aq) \rightarrow MnO_2(s) + NO_3^-(aq)$ *{Basic medium]*

18. Balance the following ionic equations. For each equation, tell what is oxidised, what is reduced, what is the oxidising agent, and what is the reducing agent.

(a) $Cr(OH)_4^{2-}(aq) + H_2O_2(aq) \rightarrow CrO_4^{2-}(aq) + H_2O(l)$ *[Basic medium]*

(b) $MnO_2(s) + NO_2^-(aq) \rightarrow NO_3^-(aq)\ Mn^{2+-}(aq) + H_2O(l)$ *[Acid medium]*

(c) $Sn(OH)_3^-(aq) + Bi(OH)_3(s) \rightarrow Sn(OH)_6^{2-}(aq) + Bi(s)$ *[Basic medium]*

(d) $CrO_4^{2-}(aq) + HSnO_2^-(aq) \rightarrow CrO_2^-(aq) + HSnO_3^-(aq)$ *[Basic medium]*

19. The iron in a *5.675* g sample containing some Fe_2O_3 is reduced to Fe^{2+}. The Fe^{2+} is titrated with *12.02* cm^3 of *0.1467* M $K_2Cr_2O_7$ in an acid solution. Find

(a) Write a balanced chemical equation for the reaction between KCr_2O_7 and Fe^{2+}

(b) the mass of Fe and

(c) The percentage of Fe in the sample.